

cleaning the old thermal paste away, and cleaning the heatsink using a cloth and some brushes between the fins. We then proceeded to connect everything back, and used the round IDE cable instead of the flat one we had earlier. We also connected the RAM strips in the slots furthest away from the CPU—you may not be allowed to do this on all motherboards—and made sure there was enough space above and below each hard drive.

We now measured the temperature again, and found that it hovered between 31 and 32 degrees celsius. That's 2 or 3 degrees cooler just because of a sound cleaning! We also found that the CPU idling temperature from the BIOS screen had dipped from 42 degrees to 38 degrees Celsius—vive le Arctic Silver!

Now our cabinet also came with a side panel with a conical appendage on the inside that's meant to help bring cool air directly to the CPU heatsink. We wanted to check whether this helped the CPU cool any better than when we attached the simple acrylic sheet. It did, but very marginally; with the stock cabinet side panel, with the conical appendage, we found that CPU temperatures stayed at 37 or 38 degrees; however, we suspect that airflow was obstructed by this cone, because cabinet temperatures were raised a degree.

For those of you wondering why a degree matters, you should remember that these are just idling temperatures, and also during the winter, in an air-conditioned office. At home, with cramped spaces within your computer table, or under a desk, while running a game or video-rendering software or something equally taxing, the temperatures of your CPU and cabinet will soar—even more so in our Indian summers, for those without air conditioning!

We decided to discard the stock side panel and stick to the acrylic sheet. Not only did it seem cooler inside the cabinet, we also needed to conduct our airflow experiments now, and needed the transparency of the acrylic.

Coughing Fits

Now it was time for the smoke. We made the mistake of trying to conduct the experiment in a closed room—bad idea. Five minutes into the whole thing, we all ran out coughing up something terrible, and found that everything, including water, seemed to taste of incense. We strongly advise that if you do this, do it in a very well-ventilated area.

Red-eyed and with weakened lungs, we bravely ventured back into the room and opened the windows, and brought along a couple of fans.

We then proceeded to fill our cabinet with smoke, and try and see how the smoke flowed. We put a few burning incense sticks (safely contained inside a ceramic plate, after one of us burnt his fingers!) into the cabinet, put the acrylic sheet on, and then waited awhile for the cabinet to fill up with smoke. We then turned the PC on and watched where the smoke went. The hardest part was trying to get



The hard disk cooler sucks in air from below and is supposed to blow it out both open sides, but for some reason, it would do so out of only one side. Go figure.



This X1300 graphics card blew air into the most inconvenient place possible—when buying, please make sure that your card's heatsink blows *out* of your cabinet!

still photographs to tell the story, but to the naked eye, the way the air flowed was obvious.

Now the fan-happy colleague we mentioned earlier had inadvertently reversed his cabinet's rear fan, causing it to draw air *in* instead of throwing it out! We did the same to our test cabinet, and noticed that this was bad for airflow, because everything was being forced out from the power supply exhaust. This left smoke in the lower regions of the cabinet for a few seconds longer. It was also opposing the way the CPU fan was trying to blow air out, causing our smoke to be dispersed wildly, instead of forming a nice neat channel of airflow. All of you should make sure that your cabinet's rear fan does indeed blow air out, because if it's acting as an input instead of an output, it could cause your cabinet to be hotter than it should be. The smoke also cleared a few other things up: